# titrations

Name: 

Class: 

Date: 

Time:

Marks: **60 marks**

Comments:

**1.** This question is about acids and alkalis.

(a) Dilute hydrochloric acid is a strong acid.

Explain why an acid can be described as both strong and dilute.

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**(2)**

(b) A $1.0 \times 10^{-3}$ mol/dm$^3$ solution of hydrochloric acid has a pH of 3.0

What is the pH of a $1.0 \times 10^{-5}$ mol/dm$^3$ solution of hydrochloric acid?

pH = ____________________

**(1)**

A student titrated 25.0 cm$^3$ portions of dilute sulfuric acid with a 0.105 mol/dm$^3$ sodium hydroxide solution.

(c) The table below shows the student's results.

| | **Titration 1** | **Titration 2** | **Titration 3** | **Titration 4** | **Titration 5** |
|---|---|---|---|---|---|
| Volume of sodium hydroxide solution in cm$^3$ | 23.50 | 21.10 | 22.10 | 22.15 | 22.15 |

The equation for the reaction is:

$$2\ NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2\ H_2O$$

Calculate the concentration of the sulfuric acid in mol/dm$^3$

Use only the student's concordant results.

Concordant results are those within 0.10 cm$^3$ of each other.

Concentration of sulfuric acid = ____________________ mol/dm$^3$

**(5)**

(d) Explain why the student should use a pipette to measure the dilute sulfuric acid and a burette to measure the sodium hydroxide solution.

**(2)**

(e) Calculate the mass of sodium hydroxide in 30.0 $cm^3$ of a 0.105 $mol/dm^3$ solution.

Relative formula mass ($M_r$): NaOH = 40

______________________________________________

______________________________________________

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Mass of sodium hydroxide = ____________________ g

**(2)**

**(Total 12 marks)**

**2.** A student investigated the temperature change in the reaction between dilute sulfuric acid and potassium hydroxide solution.

This is the method used.

1. Measure 25.0 $cm^3$ potassium hydroxide solution into a polystyrene cup.
2. Record the temperature of the solution.
3. Add 2.0 $cm^3$ dilute sulfuric acid.
4. Stir the solution.
5. Record the temperature of the solution.
6. Repeat steps 3 to 5 until a total of 20.0 $cm^3$ dilute sulfuric acid has been added.

(a) Suggest why the student used a polystyrene cup rather than a glass beaker for the reaction.

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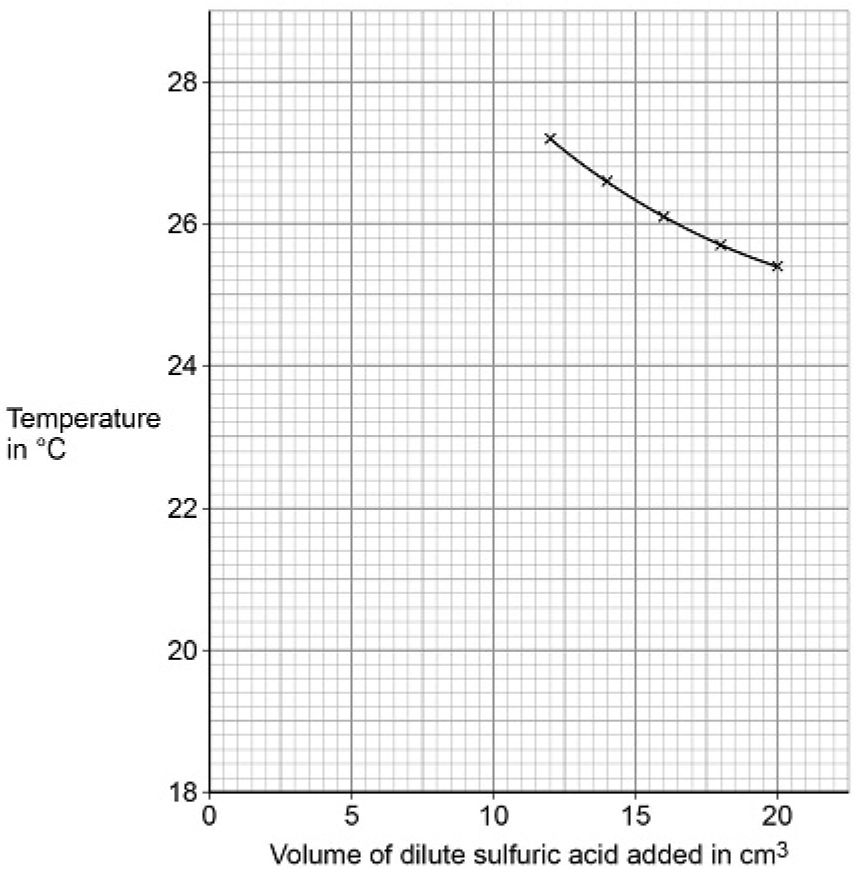
**(2)**

The following table shows some of the student's results.

| **Volume of dilute sulfuric acid added in $cm^3$** | **Temperature in °C** |
|---|---|
| 0.0 | 18.9 |
| 2.0 | 21.7 |
| 4.0 | 23.6 |
| 6.0 | 25.0 |
| 8.0 | 26.1 |
| 10.0 | 27.1 |

The figure below shows some of the data from the investigation.




(b) Complete the figure:

- plot the data from the table
- draw a line of best fit through these points
- extend the lines of best fit until they cross.

**(4)**

(c) Determine the volume of dilute sulfuric acid needed to react completely with 25.0 $cm^3$ of the potassium hydroxide solution.

Use the figure above.

Volume of dilute sulfuric acid to react completely = ______________________ $cm^3$

**(1)**

(d) Determine the overall temperature change when the reaction is complete.

Use the figure above.

______________________________________________________________________

______________________________________________________________________

Overall temperature change = __________________________ °C

**(1)**

(e) The student repeated the investigation.

The student used solutions that had different concentrations from the first investigation.

The student found that 15.5 $cm^3$ of 0.500 mol/$dm^3$ dilute sulfuric acid completely reacted with 25.0 $cm^3$ of potassium hydroxide solution.

The equation for the reaction is:

$$2\ KOH + H_2SO_4 \rightarrow K_2SO_4 + 2\ H_2O$$

Calculate the concentration of the potassium hydroxide solution in mol/dm$^3$ and in g/dm$^3$

Relative atomic masses ($A_r$): H = 1 O = 16 K = 39

Concentration in mol/dm$^3$ = ____________________ mol/dm$^3$

Concentration in g/dm$^3$ = _____________________ g/dm$^3$

**(6)**

**(Total 14 marks)**

**3.** Citric acid is a weak acid.

(a) Explain what is meant by a weak acid.

**(2)**

A student titrated citric acid with sodium hydroxide solution.

This is the method used.

1. Pipette 25.0 $cm^3$ of sodium hydroxide solution into a conical flask.
2. Add a few drops of thymol blue indicator to the sodium hydroxide solution.
   Thymol blue is blue in alkali and yellow in acid.
3. Add citric acid solution from a burette until the end-point was reached.

(b) Explain what would happen at the end-point of this titration.

Refer to the acid, the alkali and the indicator in your answer.

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**(3)**

(c) Explain why a pipette is used to measure the sodium hydroxide solution but a burette is used to measure the citric acid solution

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______________________________________________________________________

**(2)**

(d) The table shows the student's results.

| | **Titration 1** | **Titration 2** | **Titration 3** | **Titration 4** | **Titration 5** |
|---|---|---|---|---|---|
| Volume of citric acid solution in $cm^3$ | 13.50 | 12.10 | 11.10 | 12.15 | 12.15 |

The equation for the reaction is:

$$C_6H_8O_7 \quad + \quad 3\ NaOH \quad \longrightarrow \quad C_6H_5O_7Na_3 \quad + \quad 3\ H_2O$$

The concentration of the sodium hydroxide was 0.102 mol / $dm^3$

Concordant results are those within 0.10 $cm^3$ of each other.

Calculate the concentration of the citric acid in mol / $dm^3$

Use only the concordant results from the table in your calculation.

You must show your working.

____________________________________________________________

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____________________________________________________________

Concentration = ____________________ mol / $dm^3$

**(5)**

**(Total 12 marks)**

**4.** Sodium hydroxide neutralises sulfuric acid.

The equation for the reaction is:

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

(a) Sulfuric acid is a strong acid.

What is meant by a strong acid?

______________________________________________________________

______________________________________________________________

______________________________________________________________

**(2)**

(b) Write the ionic equation for this neutralisation reaction. Include state symbols.

______________________________________________________________

**(2)**

(c) A student used a pipette to add 25.0 $cm^3$ of sodium hydroxide of unknown concentration to a conical flask.

The student carried out a titration to find out the volume of 0.100 mol / $dm^3$ sulfuric acid needed to neutralise the sodium hydroxide.

Describe how the student would complete the titration.

You should name a suitable indicator and give the colour change that would be seen.

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**(4)**

(d) The student carried out five titrations. Her results are shown in the table below.

| | Titration 1 | Titration 2 | Titration 3 | Titration 4 | Titration 5 |
|---|---|---|---|---|---|
| Volume of 0.100 mol / $dm^3$ sulfuric acid in $cm^3$ | 27.40 | 28.15 | 27.05 | 27.15 | 27.15 |

Concordant results are within 0.10 $cm^3$ of each other.

Use the student's concordant results to work out the mean volume of 0.100 mol / $dm^3$ sulfuric acid added.

Mean volume = ______________________________ $cm^3$

**(2)**

(e) The equation for the reaction is:

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

Calculate the concentration of the sodium hydroxide.

Give your answer to three significant figures.

Concentration = ________________________ mol / $dm^3$

**(4)**

(f) The student did another experiment using 20 $cm^3$ of sodium hydroxide solution with a concentration of 0.18 mol / $dm^3$.

Relative formula mass ($M_r$) of NaOH = 40

Calculate the mass of sodium hydroxide in 20 $cm^3$ of this solution.

___

___

___

___

Mass = ________________ g

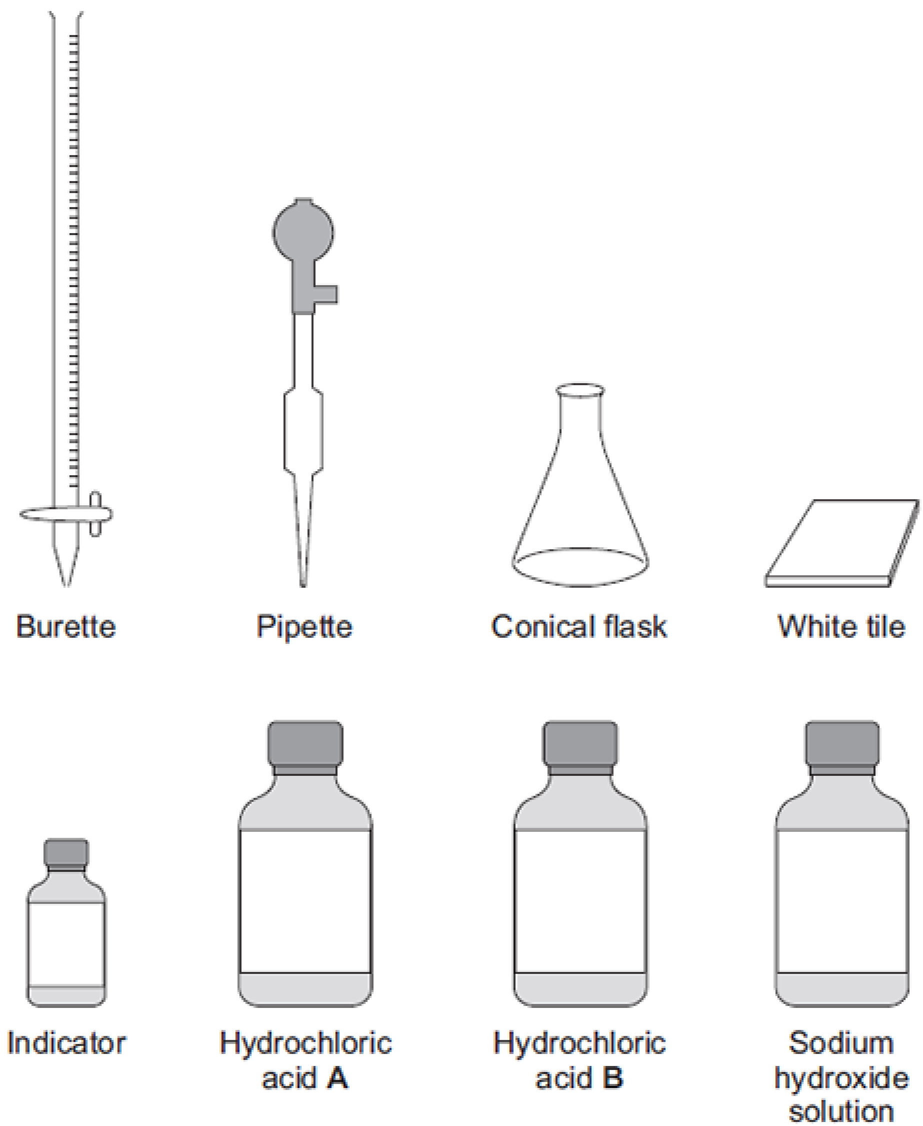
**(2)**

**(Total 16 marks)**

**5.** **In this question you will be assessed on using good English, organising information clearly and using specialist terms where appropriate.**

A student has to check if two samples of hydrochloric acid, **A** and **B**, are the same concentration.

Describe how the student could use the apparatus and the solutions in the diagram below to carry out titrations.




**(Total 6 marks)**

## Mark schemes

**1.** (a) (strong because) completely ionised (in aqueous solution)

*ignore pH*

*allow dissociated for ionised*

*do **not** accept hydrogen is ionising*

*do **not** accept $H^+$ are ionised*

**1**

(dilute because) small amount of acid per unit volume

*ignore low concentration*

**1**

(b) 5.0

*allow 5*

**1**

(c) (titre):

chooses titrations 3, 4, 5

1

average titre = 22.13 ($cm^3$)

*allow average titre = 22.13(3…) ($cm^3$)*
*allow a correctly calculated average from an incorrect choice of titrations*

1

(calculation):
(moles NaOH =
$\frac{22.13}{1000} \times 0.105 = 0.002324$)

*allow use of incorrect average titre from step 2*

1

(moles $H_2SO_4$ =
½ × 0.002324 =) 0.001162

*allow use of incorrect number of moles from step 3*

1

(concentration =
$\frac{0.001162}{25} \times 1000$)

= 0.0465 ($mol/dm^3$)

*allow use of incorrect number of moles from step 4*

1

*alternative approach for step 3, step 4 and step 5*

$$\frac{2}{1} = \frac{22.13 \times 0.105}{25.0 \times conc.\ H_2SO_4} \ (1)$$

*(concentration $H_2SO_4$ =)*

$$\frac{22.13 \times 0.105}{25.0 \times 2}$$

*= 0.0465 ($mol/dm^3$) (1)*

*an answer of 0.046473* ***or*** *0.04648 correctly rounded to at least 2 sig figs scores marking points 3, 4 and 5*

*an answer of 0.092946* ***or*** *0.09296* ***or*** *0.185892* ***or*** *0.18592 correctly rounded to at least 2 sig figs scores marking points 3 and 5*

*an incorrect answer for one step does* ***not*** *prevent allocation of marks for subsequent steps*

(d) pipette measures a fixed volume (accurately)

1

(but) burette measures variable volume

*allow can measure drop by drop*

1

(e) $(\text{moles} =) \dfrac{30}{1000} \times 0.105$

**or** 0.00315 (mol)

**or**

(mass per dm$^3$ =) 0.105 × 40

**or** 4.2 (g)

1

$(\text{mass} = \dfrac{30}{1000} \times 0.105 \times 40)$

= 0.126 (g)

1

*an answer of 0.126 (g) scores **2** marks*

*an answer of 126(g) scores **1** mark*

*an incorrect answer for one step does **not** prevent allocation of marks for subsequent steps*

**[12]**

**2.** (a) polystyrene is a better (thermal) insulator

*allow polystyrene is a poorer (thermal) conductor*

1

(so) reduces energy exchange (with the surroundings)

*allow (so) reduces energy / heat loss (to the surroundings)*

1

(b) all six points plotted correctly

*allow a tolerance of ± ½ a small square*

*allow **1** mark for at least 3 points plotted correctly*

2

line of best fit through points plotted from the table

1

both lines of best fit extrapolated correctly until they cross

1

(c) 11 (cm$^3$)

*allow ecf from part (b)*

*allow answers in the range 10.75 to 11.25 (cm$^3$)*

*allow a tolerance of ± ½ a small square*

1

(d) (27.5 − 18.9) = 8.6 (°C)

*allow ecf from part (b)*

*allow answers in the range 8.5 to 8.7 (°C)*

*allow a tolerance of ± ½ a small square*

1

(e)

*an answer of 0.62 (mol/dm$^3$) for concentration in mol/dm$^3$ scores **4** marks*

*an answer of 0.31 (mol/dm$^3$) for concentration in mol/dm$^3$ scores **3** marks*

(moles $H_2SO_4$ = 0.500 × $\frac{15.5}{1000}$) = 0.00775

**1**

(moles KOH = 2 x moles $H_2SO_4$ = 2 x 0.00775) = 0.0155

*allow correct calculation using incorrectly calculated value of moles of $H_2SO_4$*

**1**

(conc KOH = moles KOH x $\frac{1000}{25.0}$ ) = 0.0155 x $\frac{1000}{25.0}$)

*allow correct calculation using incorrectly calculated value of moles of KOH*

**1**

= 0.62 (mol/dm$^3$)

*allow correct answer using incorrectly calculated value of moles of KOH*

**1**

($M_r$ KOH =) 56

**1**

(conc = $M_r$ x conc in mol/dm$^3$ = 56 x 0.62) = 34.7 (g/dm$^3$)

*allow 35 or 34.72 (g/dm$^3$)*

*allow correct answer using incorrectly calculated value of concentration in mol/dm$^3$ and/or incorrect $M_r$*

**1**

**alternative approach for step 1 to step 4**

$\frac{2}{1} = \frac{25 \times \text{conc KOH}}{15.5 \times 0.500}$ (2)

(conc KOH) = $\frac{2 \times 15.5 \times 0.500}{25.0}$ (1)

= 0.62 (mol/dm$^3$) (1)

*allow 1 mark if mole ratio is incorrect*

**1**

**[14]**

**3.** (a) produces $H^+$ / hydrogen ions in aqueous solution

**1**

(but is) only partially / slightly ionised

**1**

(b) indicator changes colour

1

from blue to yellow

*allow from blue to green*

1

(when) the acid and alkali are (exactly) neutralised
**or**
(when) no excess of either acid or alkali

1

(c) pipette measures one fixed volume (accurately)

1

(but) burette measures variable volumes (accurately)

1

(d) $\frac{12.10 + 12.15 + 12.15}{3}$

1

(mean titre =) 12.13(3) ($cm^3$)

1

(moles NaOH = conc × vol) = 0.00255

1

(moles citric acid = $\frac{1}{3}$ moles NaOH) = 0.00085

1

(conc acid = moles / vol) = 0.0701 (mol / $dm^3$)

*allow ecf from steps 1, 2, 3 and / or 4*

*allow an answer of 0.0701 (mol / $dm^3$) without working for **1** mark only*

1

**[12]**

**4.** (a) (sulfuric acid is) completely / fully ionised

1

In aqueous solution **or** when dissolved in water

1

(b) $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

*allow multiples*

***1** mark for equation*

***1** mark for state symbols*

2

(c) adds indicator, eg phenolpthalein / methyl orange / litmus added to the sodium hydroxide (in the conical flask)

*do **not** accept universal indicator*

1

(adds the acid from a) burette

1

with swirling **or** dropwise towards the end point **or** until the indicator just changes colour

1

until the indicator changes from pink to colourless (for phenolphthalein) or yellow to red (for methyl orange) or blue to red (for litmus)

1

(d) titrations 3, 4 and 5

**or**

$$\frac{27.05 + 27.15 + 27.15}{3}$$

1

27.12 $cm^3$

*accept 27.12 with no working shown for **2** marks*

1

*allow 27.1166 with no working shown for **2** marks*

(e) Moles $H_2SO_4$ = conc × vol = 0.00271

*allow ecf from 8.4*

1

Ratio $H_2SO_4$:NaOH is 1:2

**or**

Moles NaOH = Moles $H_2SO_4$ × 2 = 0.00542

1

Concentration NaOH = mol / vol = 0.00542 / 0.025 = 0.2168

1

0.217 (mol / $dm^3$)

*accept 0.217 with no working for **4** marks*

1

*accept 0.2168 with no working for **3** marks*

(f) $\frac{20}{1000}$ × 0.18 = no of moles

**or**

0.15 × 40 g

1

0.144 (g)

1

*accept 0.144g with no working for **2** marks*

**[16]**

**5.** Marks awarded for this answer will be determined by the Quality of Written Communication (QWC) as well as the standard of the scientific response. Examiners should also apply a ‘best-fit’ approach to the marking.

**Level 3 (5 – 6 marks)**
There is a description of titrations that would allow a comparison to be made between the two solutions of hydrochloric acid.

**Level 2 (3 – 4 marks)**
There is a description of an experimental method including addition of acid to alkali which may include an indicator or colour change and may include a measurement of volume.

**Level 1 (1 – 2 marks)**
There is a simple description of using some of the apparatus.

**0 marks**
No relevant content.

**examples of chemistry points made in the response could include:**
- acid in burette **or** flask
- alkali/sodium hydroxide **or** acid in burette **or** flask
- volume of acid **or** alkali measured using the pipette
- indicator in flask
- white tile under the flask
- slow addition
- swirling/mixing
- colour change of indicator
- burette volume measured

**[6]**

## Examiner reports

**1.** (a) 46% of students could explain the idea of a strong acid correctly, but few could articulate the idea of a dilute acid using correct scientific ideas. Most referred either to a low concentration (which merely expresses the opposite of the word dilute) or adding water, explaining the word diluted rather than dilute. To gain the mark, the idea of few particles per unit volume was required. Some students tried to explain the ideas in terms of pH, failing to realise that pH is itself affected by both the strength of an acid and by its concentration.

(b) This question, on a new area of the specification, was answered well, with 68% of students giving the correct answer.

(c) This question discriminated very well. The highest-attaining students gave excellent answers, with clearly set out working which was easy to follow. 20% of students achieved full marks, and 41% were able to achieve at least two marks, usually for calculating the correct average titre from concordant results.

Some working was very hard to follow, making it difficult to give partial marks. This was particularly the case when students wrote numbers around a pair of triangles rather than giving a step by step process. It is as important to communicate how the answer is obtained as it is to get an answer. It is also hard to follow working when partial answers are left as fractions bearing no relation to the actual data; a calculator display should always be converted to decimal form for intermediate steps as well as the final answer.

Some students had a correct answer but then took their final answer down to one significant figure. Having done a required practical, students should know that titration is a very accurate technique so that an answer to one significant figure would never be appropriate.

(d) There was little understanding of the reasons for using different apparatus to measure the two volumes. Answers often did not mention volume at all but referred vaguely to ‘amount’ or ‘how much’. Both a pipette and a burette measure accurately and precisely − the difference is that a pipette can only measure one single volume, but a burette measures variable volumes.

(e) 32% of students achieved both marks. The most common errors were to forget to divide by 1000 when calculating the number of moles, or to calculate the mass per dm3 but not then scale to 30 $cm^3$.

**2.** (a) Almost a half of students scored 0 marks.

The mark scheme required the use of comparatives.

Common correct responses were:

- Better insulator
- Less heat loss

Common responses seen that were not credited included:

- good insulator
- no heat loss.

(b) This graph question was well answered with approximately half of students scoring full marks.

Some students attempted to draw a straight line of best fit. In most cases, this was inappropriate.

Some students did not draw their line so that the two lines of best fit crossed. Instead, they drew a curved peak.

(c) Approximately 60% of students were able to determine the neutralisation point from their graph.

In some cases, students misunderstood the scale of the x-axis, for example, reading 11 $cm^3$ as 10.1 $cm^3$.

(d) Only approximately a third of students correctly determined the overall temperature change for the reaction from their graph.

A common incorrect answer was 8.2 °C (from subtraction of the highest and lowest temperature values in the table).

(e) Students found this titration calculation difficult and approximately a third of students scored 0 marks.

The marks for calculating the number of moles of sulfuric acid and for calculating the relative formula mass of potassium hydroxide were the most scored. Even when these values were incorrect, this did not prevent students from gaining marks in subsequent steps.

The correct answer for the concentration in mol/dm3 was achieved more often than the subsequent conversion to g/$dm^3$. A common error was to use a conversion factor of 112 (2 × $M_r$(KOH)) rather than 56.

**5.** **Foundation**

This six-mark question contained the command word ‘describe’. It produced an even spread of marks. Some high quality Level 3 responses were seen which described a titration procedure that would enable a valid comparison of the two acid samples. There were some descriptions of features which would improve accuracy. Weaker responses may not have stated or implied that the procedure should be applied to both acid samples, which confined them to Level 2. Other misconceptions were that titrations can be successfully carried out by counting drops from the burette, that acid-base indicators turn clear, or that Universal Indicator could just be used directly on the acid samples and a comparison of pH made. A few students became confused with thiosulfate rate experiments they may have done, and therefore drew crosses on the white tile to become obscured (or visible) when the indicator changed colour. Most students were at least able to describe an addition of acid to alkali or vice-versa, and so gained a Level 2 mark or above, although some mistakenly added acid A to acid B.

A high standard of written communication was evident in some responses. Some students, as well as failing to organise their response in a logical or coherent way, tended to show particular problems in the spelling of apparatus words such as ‘burette’, ‘pipette’ and ‘conical’ (even though they were given in the question stem) and reagent and indicator names, and used everyday words rather than scientific terms.

### Higher

This six-mark question contained the command word ‘describe’. Many high quality Level 3 responses were seen which described a titration procedure that would enable a valid comparison of the two acid samples. There were many descriptions of features which would improve accuracy. Weaker responses may not have stated or implied that the procedure should be applied to both acid samples, which confined them to Level 2. Other misconceptions were that titrations can be successfully carried out by counting drops from the burette, that acid-base indicators turn clear, or that Universal Indicator could just be used directly on the acid samples and a comparison of pH made. A few students became confused with thiosulfate rate experiments they may have done, and therefore drew crosses on the white tile to become obscured (or visible) when the indicator changed colour. Nearly all students were at least able to describe an addition of acid to alkali or vice-versa, so the number of Level 1 responses was pleasingly low.

A high standard of written communication was evident in many responses. Some students, as well as not organising their response in a logical or coherent way, tended to show particular problems in the spelling of apparatus words such as ‘burette’, ‘pipette’ and ‘conical’ (even though they were given in the question stem) and reagent and indicator names, and used everyday words rather than scientific terms.